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WORK EXPERIENCE

Software Engineer, Creative Medtech Solutions

Aug 2023 – Present

- Built and maintained software features in Qt and C++ for UroMaster, a marquee robotic fusion biopsy system.
- Streamlined the Scrum-based software development lifecycle by creating CI workflows and automating testing processes using GoogleTest.
- Devised a non-intrusive, generic, extensible and customizable logging wrapper for Qt Widgets, fulfilling HSA cybersecurity requirement. Wrote more than 50 unit and integration tests.
- Implemented automatic screen rotation mechanism with Qt and rewrote using Windows API to overcome Qt limitations and support kiosk mode. Enabled market release without requiring HSA reapprovals, saving \$560 dollars per document.

Software Engineer Intern, Omron Asia Pacific Pte Ltd

Jan 2022 – Jun 2022

- Worked on Omron's advanced research and development team, created, enhanced and shipped applications employing Omron industrial robots.
- Conceptualized and developed a solution for autonomous mobile robot (AMR) cart alignment using OpenCV-based ArUco marker detection and motion planning. Collaborated with a mechanical engineer to oversee hardware fabrication and assembly. Adopted and improved by a client for factory use.
- Invented a gripper control procedure using proprietary software to substitute for missing adapter in 2 days, enabling a live mobile pick-and-place demo to a delegation from a massive food manufacturer. Client agreed to further production line experiments.
- Programmed and presented a prototype integration of in-house AMR control platform with open-sourced Robot Middleware Framework in 3 weeks. Prepared documentation to streamline development flow for next interns.

SKILLS

Programming Languages: Go, C++, Python

Databases: PostgreSQL, SQLite

Cloud Platforms: Heroku, Railway

Methodologies: Agile, Scrum, Iterative, Waterfall

EDUCATION

Bachelor of Computing (Hons), Computer Science, National University of Singapore

Aug 2019 – Jun 2023

- Artificial Intelligence focus area

PROJECTS

Snippetbox, Sole Developer

Sep 2024 – Present

- Developed and tested an application for saving and viewing text snippets with built-in user authentication using Go, PostgreSQL with basic HTML and CSS.
- Constructed a CI/CD workflow to automate code linting, vulnerability checking, testing and deployment to Railway, reducing test duration by 15% by using faster runners.

Image-based Threat Detection, Team Lead

Oct 2022 – Nov 2022

- Led a team of 3 in a computer vision course project.
- Coordinated manual data inspection process, formulated and programmed a pluggable data preprocessing pipeline for more than 4000 images.
- Classified images into different threat levels using transfer learning on ResNet and Inception models with TensorFlow, resulting in respectable accuracy of 78% on a dataset of inconsistent quality.
- Visualized results and aided poster design stage, presented and clarified questions from 4 random teaching team members.